

Transistor

The **COMPACT COMPONENT** that made modern electronics possible



Vacuum tubes

Transistors are used in electronic equipment to **switch or amplify** electric signals. Before transistors, these jobs were done by vacuum tubes, which looked like light bulbs and were **unreliable and bulky**. American physicists William Shockley, Walter Brattain, and John Bardeen began developing ideas to **REPLACE** them in the 1950s.

How it changed

Transistors made electronic equipment smaller and more reliable. Without them, the gadgets we use every day wouldn't exist.

the world

Brattain once said, "The only regret I have about the transistor is its use for rock and roll."

The three leads allow the transistor to stop, start, or increase electrical current.

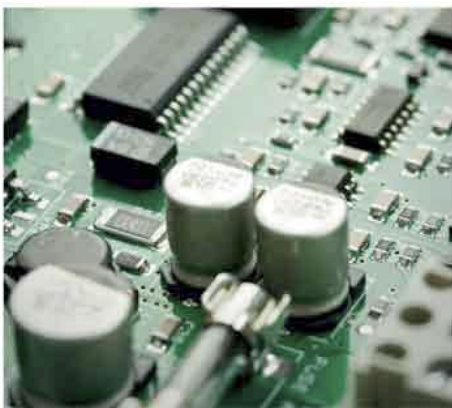
Transistors take over

The trio's small but revolutionary solution, called the **TRANSISTOR**, could control electric current just like vacuum tubes did, but was an enormous improvement. The transistor used far less power, hardly ever failed, and was so tiny that it made it possible to have **smaller electronic equipment**.

In 1956, Brattain, Bardeen, and Shockley were jointly awarded the **Nobel Prize for Physics** for their work on transistors.

Shockley worked for the US military during World War II.

Bardeen went on to win a second Nobel Prize for Physics in 1972.



Circuitry

Early transistors were about the length of the **palm of your hand**, but improvements to their design led them to become **SMALLER**. At first, they were connected to other electronic components on circuit boards, and used in hearing aids, radios, and computers. Now, transistors are mostly found in computer chips—**hundreds of millions** of transistors can fit on a single chip.